

SUFFOLK COUNTY COMMUNITY COLLEGE
ANNUAL REPORT OF ACADEMIC ASSESSMENT ACTIVITY

Thank you for completing the academic assessment activity report. This documentation is important in demonstrating our commitment to using student learning assessment to improve teaching and learning.

Academic Year reporting on: Year # in 5-Year Assessment Plan: **Fall 2025 - Spring 2026**

There are three sections to this report:

Section I: Report on assessment activity follow-up from the previous Academic Year:

Section II: Report on current assessment activity for this Academic Year:

Section III: Report on Program Review Recommendations.

ANNUAL REPORT OF ACADEMIC ASSESSMENT ACTIVITY

Academic Program and Degree:	Liberal Arts and Sciences: Science – Earth and Space Science – Astronomy Option (LAAS-AS)		
Lead person(s):	Glenda Denicolo		
Date of Assessment Report:	May 29, 2026		
Does this PO align with a SUNY General Education learning outcome?	If YES, please indicate which one: The PO assessed in this report is PLO#1 (<i>“Demonstrate knowledge of factual material essential to their discipline in science.”</i>), which aligns with the following SUNY GE learning outcomes: <i>Mathematics:</i> ○ <i>interpret and draw inferences from appropriate mathematical models such as formulas, graphs, tables, or schematics;</i> ○ <i>represent mathematical information symbolically, visually, numerically, or verbally as appropriate; and</i> ○ <i>employ quantitative methods such as arithmetic, algebra, geometry, or statistics to solve problems.</i> <i>Natural Sciences:</i> ○ <i>an understanding of the methods scientists use to explore natural phenomena, including observation, hypothesis development, measurement and data collection, experimentation, evaluation of evidence, and employment of data analysis or mathematical modeling; and</i> ○ <i>application of scientific data, concepts, and models in one of the natural sciences.</i> <i>Critical Thinking and Reasoning:</i> ○ <i>clearly articulate an issue or problem;</i> ○ <i>identify, analyze, and evaluate ideas, data, and arguments as they occur in their own or others’ work;</i> ○ <i>acknowledge limitations such as perspective and bias; and</i> ○ <i>develop well-reasoned (logical) arguments to form judgments and/or draw conclusions.</i>		

Section I: Report on assessment activity follow-up from Academic Year Fall 2024/Spring 2025.

What Program Learning Outcome(s) did you assess last academic year?	Program Learning Outcome #1: <i>Demonstrate knowledge of factual material essential to their discipline in science.</i>
Describe any of actions that were taken this year in response to the assessment findings. Please give specific examples of any changes you have made as a result of the findings.	Last year's assessment report was broadly shared, which certainly helped to increase the participation this current year. We have also consistently reached out to faculty a few weeks before the start of the semester to invite participation. As a result, from an initial sample of 91 students during F24-S25, we now have a sample of 266 students for the year F25-S26. This is a successful achievement. Some adjustment to laboratory instructions have been implemented by G. Denicolo due to the results from the ADT question-by-question analysis reported last year. Question 10 scored the lowest in last year's sample, so an attempt has been made to increase exercises regarding the apparent motion of the Sun in the sky.

Section II:

Report on completed assessment activity for Current Academic Year: Fall 2025/Spring 2026.

What Program Learning Outcome(s) did you assess?	Program Learning Outcome #1: <i>Demonstrate knowledge of factual material essential to their discipline in science.</i>
Description of assessment activity. Please include (1) the methodology, as well as any specific measurement criteria; (2) the course(s) the assessment took place. (3) the desired student performance from (4) the number of students who participated in the assessment.	We have applied the research-based assessment tool called Astronomy Diagnostic Test 2.0 (ADT) to a total of sixteen sections of AST101 (11 sections, 195 students) and AST102 (5 sections, 71 students) in the past academic year. Both AST101 and AST102 received the same questions, the same assessment instrument. ADT, its resources and references can be found at https://www.physport.org/assessments/. The content knowledge covered in ADT includes mainly: apparent motion of the Sun, scale of the solar system, phases of the moon, linear distance scales, seasons, global warming, nature of light, gravity, stars, cosmology. Information on how to obtain the test is attached at the end. (1) Methodology: ADT was applied on the first day of classes (pre-test) and on the last week of the semester (post-test). Students were given approximately 30 minutes to complete the test, but most students completed it faster than that. Only 21 questions relate to the subject of astronomy, making 21 as the maximum score. The remaining questions addressed confidence level, highest math-level taken, ethnicity, and course expectations. This report is only about the first 21 questions. (2) Courses where ADT was applied this academic year: All courses were traditional-style in-person courses, except for AST101 CRN92874 and AST101 CRN 33452, which were hybrid or also called

blended courses (remote online lecture combined with in-person laboratory activities).

Semester	Course - CRN	Number of students matching Pre/Post tests	Instructor (Campus)
Fall 2025	AST101 - 90833	20	Glenda Denicolo (A)
	AST101 - 95844	19	Glenda Denicolo (A)
	AST102 - 95204	10	Glenda Denicolo (A)
	AST101 - 92201	14	Philip Harrington (A)
	AST101 - 92866	21	Mathew Pappas (A)
	AST101 - 92874 <i>hybrid</i>	19	Mathew Pappas (A)
Spring 2026	AST101 - 20743	15	Glenda Denicolo (A)
	AST101 - 40080	14	Glenda Denicolo (A)
	AST102 - 33955	11	Glenda Denicolo (A)
	AST102 - 28661	12	Philip Harrington (A)
	AST102 - 21581	19	Darin Mihalik (G)
	AST102 - 36339	19	Darin Mihalik (G)
	AST101 - 39934	15	Mary Griffin (E)
	AST101 - 35522	21	Mathew Pappas (A)
	AST101 - 39146	20	Mathew Pappas (A)
	AST101 - 33452 <i>hybrid</i>	17	Mathew Pappas (A)
Total:		266	

(3) Desired student performance:

Our objective is to compare our results with ADT national statistics presented below for the *Lecture + lab* format:

	Lecture	Lecture + discussion	Lecture + lab	Lecture + lab + discussion
# classes	16	5	40	9
# students pretest	1045	549	1730	723
Mean prescore (21 max)	6.65	6.09	6.70	7.31
Standard deviation	.71	.28	1.09	.90
# students posttest	758	369	1371	582
Mean postscore (21 max)	9.66	8.60	9.77	11.44
Standard deviation	1.61	1.13	1.88	2.01
Mean normalized gain	.2098	.1681	.2146	.3016
Standard deviation on normalized gain	.0870	.0628	.0973	.1221
Standard error on normalized gain	.0217	.0281	.0154	.0407

Source: Brogt et al., 2007. <https://doi.org/10.3847/AER2007003>

Database: <https://solar.physics.montana.edu/aae/adt/>

Following the literature, we have also used normalized gain (g) and effect size (d) as metrics for comparison.

$$\text{Normalized gain: } g = \frac{\langle \text{post} \rangle - \langle \text{pre} \rangle}{(\text{max score}) - \langle \text{pre} \rangle}$$

<https://www.physport.org/recommendations/Entry.cfm?ID=93334>

	The effect size: $d = \frac{<post> - <pre>}{std\ dev}$ https://www.physport.org/recommendations/Entry.cfm?ID=93385 (4) Total number of students who participated in the assessment this year: 266 students (195 from AST101, 71 from AST102).
--	---

Summary and interpretation of the findings. What evidence is there that the learning outcome is being met?

This is the second year we are using ADT to assess general knowledge in astronomy (PLO#1). Our goal is to compare our results with published national data and study the differences. The table below summarizes our comparative results this year (the uncertainties are standard deviations):

Table 1 – Summary of comparative results with literature.

F25-S26	SCCC result	Brogt et al 2007	Comparisons
Total number of students for lab+lecture format	266 matched	Pre: 1730 Post: 1371	⚠ caution
Mean Pre-score	6.04 ± 3.05	6.70 ± 1.09	✓ OK
Mean Post-score	7.24 ± 3.54	9.77 ± 1.88	✓ OK
Mean Normalized Gain, g	0.01 ± 0.01	0.21 ± 0.10	⚠ caution
Mean Effect Size, d	0.36 ± 0.26 (medium)	Not available*	

* The data in the paper *Brogt et al., 2007* does not contain samples matched student-by-student, thus no effect size d can be directly calculated.

Table 2 – Detailed values of this year's assessment results using ADT per course section.

F25-S26 AST-CRN	Sample size	g	d	Mean Pre-score	Mean Post-score
101-90833	20	0.02	0.66	5.2	6.85
101-95844	19	0.02	0.74	4.9	7.1
101-92201	14	0.00	0.00	6.5	6.5
101-92866	21	0.01	0.30	6.1	7.2
101-20743	15	0.01	0.14	6.4	7.0
101-40080	14	0.02	0.44	5.7	7.8
101-39934	15	0.01	0.25	6.9	7.9
101-35522	21	0.01	0.29	6.6	7.9
101-39146	20	< 0.01	0.17	6.5	7.0
101-92874 hybrid	19	0.00	0.08	5.8	6.2
101-33452 hybrid	17	0.02	0.57	5.4	6.8
102-95204	10	0.02	0.66	5.9	7.7
102-33955	11	0.01	0.42	7.2	8.4
102-28661	12	0.00	0.02	5.7	5.8
102-21581	19	0.02	0.44	6.3	8.2
102-36339	19	0.01	0.50	6.3	7.9

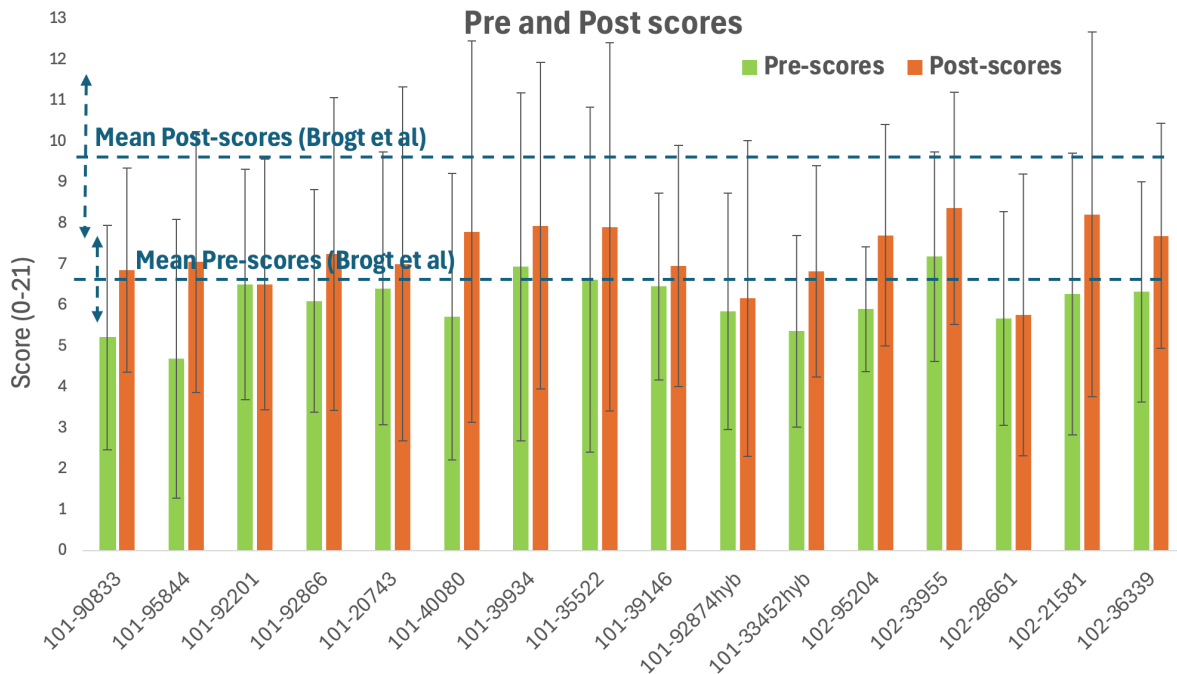


Figure 1 – Pre and Post scores (green and orange, respectively) per course section. The dashed lines represent the averages in the work of Brogt et al (2007). Error bars are standard deviations.

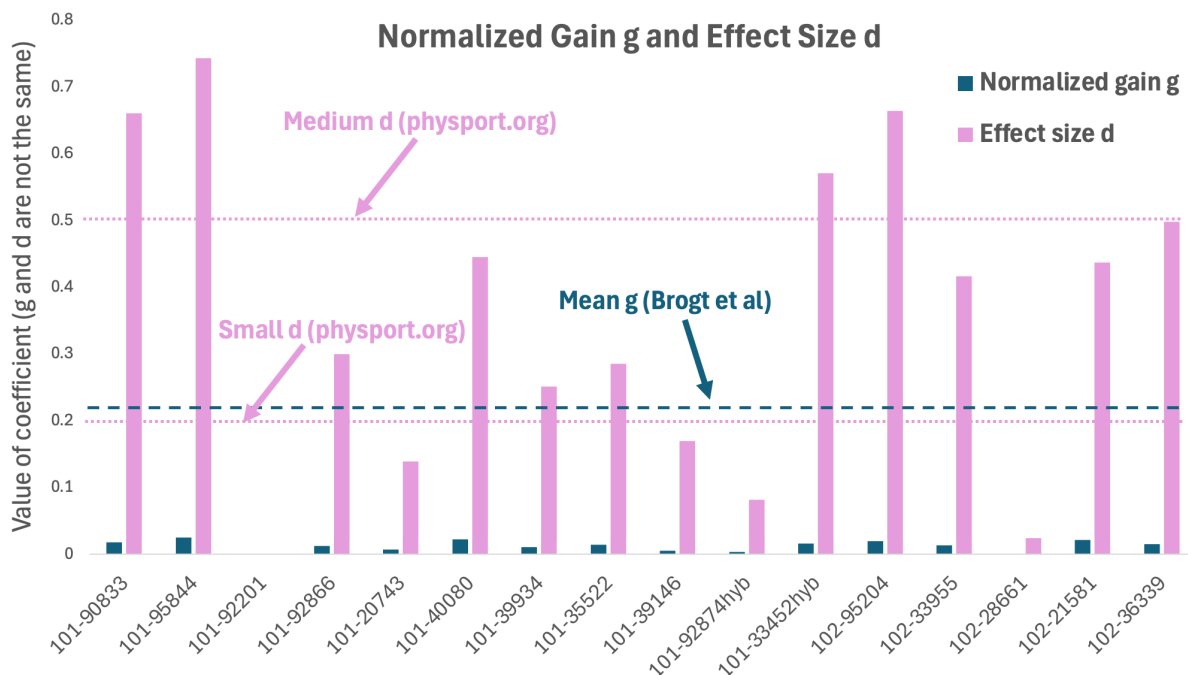


Figure 2 – Normalized gain g (dark blue, small values) and effect size d (light pink) per course section. The blue dashed line represents the average g presented in Brogt et al (2007). The pink dotted lines mark the consensus for effect size values classified as small and medium (large effect size is 0.8 and above).

Evidence that the learning outcome is being met:

- Our **average Pre- and Post-scores** (Pre: 6.0 ± 3.0 vs 6.7 ± 1.1 , and Post: 7.2 ± 3.5 vs 9.8 ± 1.9) are **within the literature values** when taking into consideration the standard deviations.
- Our effect size $d = 0.36 \pm 0.26$ is in the range of **small-to-medium** values (between 0.2 and 0.5). Effect size measures the impact in our student's learning, and is an index known to be more reliable for smaller samples like ours, as opposed to normalized gain g . **Our sample is only 20% or less the size of the sample in Brogt et al.** Overall, we interpret our value of d for the current larger sample of students and varied instructors as a **reasonable result**, but would like to improve it in the future.
- **A one-to-one comparison between our sample and data in Brogt et al is not possible.** A small fraction of the sample of students in Brogt et al comes from two-year institutions, whereas the majority are 4-year college or university students. Community colleges typically enroll more first-generation students than 4-year institutions, also, more part-time students, broader demographics, returning adult students, more academically underprepared students, more commuters and more students balancing work/family obligations. Thus, **our samples differ in math preparation, scientific reasoning preparation, academic confidence and any prior exposure to science.** It is typical to find **greater standard deviation in assessment tests from a sample of community college students** than 4-year university students. Even though our sample is only a fraction of the sample in Brogt et al, we think our large standard deviations will remain typical to our type of institution. Last but not least, **most courses in the database used by Brogt et al are single-semester astronomy courses, rather than the 2-semester sequence of courses as we have at SCCC** (AST101 and AST102). Thus, ADT's content coverage is split between two courses for us at SCCC. We will address this difference next.

The ADT data collected can also give us insight about the uniformity of content coverage and instructional methods applied in both AST101 and AST102 at our school. According to Hufnagel (2002), the questions in ADT are split into 10 categories (Table 3). We have attempted to classify the coverage of these topics by AST101 and AST102 using a 0-2 scale:

Coverage classification:

2 – dedicated unit/module or repeated instruction; appears on exams/homework

1 – brief mention or supporting concept only

0 – not meaningfully addressed

Table 3 – The ten topics covered in ADT.

ADT topics <i>Hufnagel B., 2002</i>	ADT questions related to each topic	Coverage in AST101	Coverage in AST102
Apparent motion of the Sun	Q1, Q9, Q10	2	1
Scale of the Solar System	Q3, Q11, Q13	2	0
Phases of the Moon	Q2, Q18, Q19	2	0
Linear distance scales	Q12, Q20	1	2
Seasons	Q7	2	0
Climate change	Q21	1	0
Nature of light	Q5, Q15	2	2
Gravity	Q4, Q6, Q14	2	2
Stars	Q8, Q17	1	2
Cosmology	Q16	0	2

The figures below are the result of this year's data collection shown question-by-question for AST101 and AST102, separately.

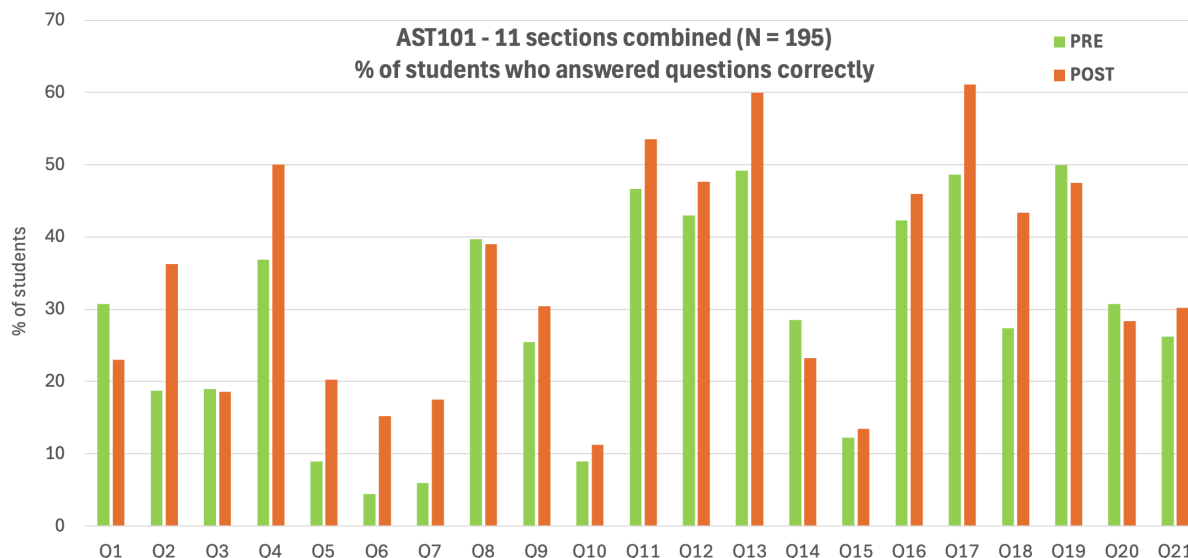


Figure 3 – Question-by-question results for all AST101 sections.

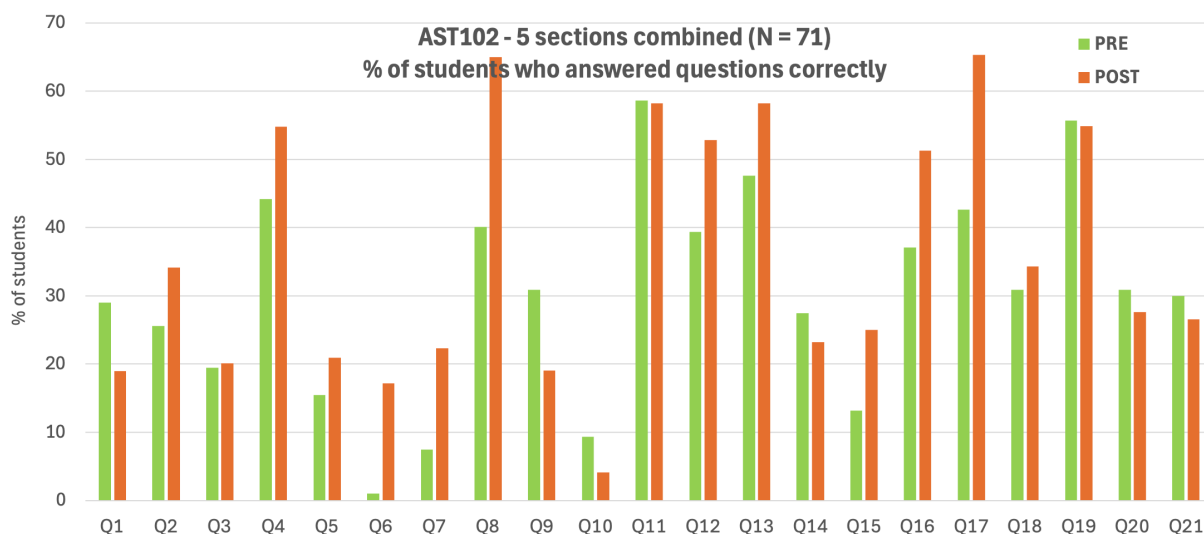


Figure 4 – Question-by-question results for all AST102 sections.

We observed that, for AST101, several questions showed no improvement in post-instruction scores (Q1, Q3, **Q8**, Q14, Q19, and Q20). Of these, only Q8 (shown in bold) is not typically associated with topics covered in AST101. The remaining questions may warrant further reflection and discussion among the faculty regarding instructional emphasis and student learning outcomes.

For AST102, the questions without improved post-scores are **Q1**, **Q9**, **Q10**, **Q11**, Q14, **Q19**, Q20, **Q21**. In this list, Q1, Q9, Q10, Q11, Q19, Q21 (shown in bold before) are not typical major topics in AST102, but Q14 and Q20 are assumed to be.

Based on these observations, we share a table with a summary of recommendations for improvements and acknowledgement of successes in both AST101 and AST102 taught at SCCC.

Table 4 – Recommendations and acknowledgements of success.

AST101		AST102	
Possible improvement	Success achieved	Possible improvement	Success achieved
Q1, Q3, Q14, Q19, Q20 are assumed to be major coverage topics in AST101, but the average student performance indicates improvements are needed.	Q2, Q4, Q11, Q13, Q17 show the largest gains when comparing pre- and post-scores. Except for Q17, all the others are major topics in AST101.	Q14, Q20 are assumed to be major coverage topics in AST102, but the average student performance indicates improvements are needed.	Q2, Q4, Q6, Q7, Q8, Q12, Q13, Q15, Q16, Q17 show the largest gains when comparing pre- and post-scores. Except for Q2, Q7, Q13, all the others are major topics in AST102.
How will assessment results be shared with program faculty, staff and other stakeholders?		The assessment report will be emailed to the astronomy faculty in the three campuses and to our department chair. Assessment results are often presented at the department meetings as well.	
Actions required to improve teaching and learning in light of the findings. Who will be responsible for overseeing any actions?		Group discussion to evaluate whether greater consistency in content coverage and instructional approaches can be achieved across AST101 and AST102. Responsible: Glenda Denicolo and Matthew Pappas.	
Description and timeline for follow-up activities.	Two weeks before start of the semester: Remote meeting with the astronomy faculty to review the results of this year's assessment and discuss the implementation of standardized assessment in the next academic year. Starting in Fall 2026, we will adopt a different research-based assessment instrument, the Test of Astronomy Standards (TOAST). Although no standardized assessment aligns perfectly with the specific topic coverage of AST101 and AST102, TOAST is a newer instrument and will provide a valuable baseline dataset, as it has not previously been administered in these courses.		

Section III: Report on Program Review Recommendations

What is the date of the last program review?	Last Program Review was <u>March 2021</u> .
Please list all of the recommendations that were made and indicate what you have done to address each recommendation and how you are making progress. If you are not making progress, explain why.	This item has been addressed in previous reports. There are no more recommendations in astronomy yet to be met.

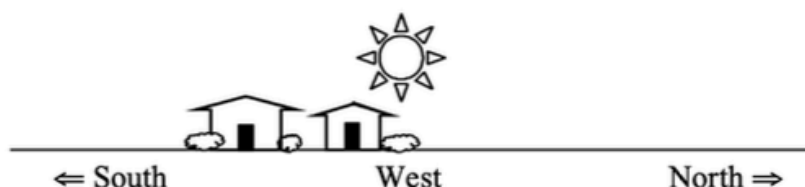
Please send any material associated with the assessment (Excel spreadsheet/ rubric/ reports, etc.) with this report to Lauren Tacke-Cushing, Associate Vice President of Academic Affairs @ tackel@sunysuffolk.edu and to your campus Associate Dean of Academic Affairs. Thank you.

Introductory Astronomy Survey

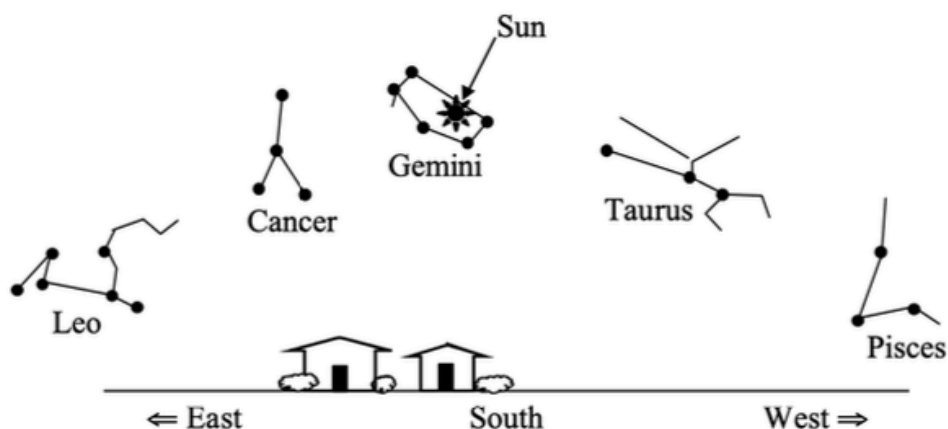
1. As seen from your current location, when will an upright flagpole cast no shadow because the Sun is directly above the flagpole?
 - A. Every day at noon.
 - B. Only on the first day of summer.
 - C. Only on the first day of winter.
 - D. On both the first days of spring and fall.
 - E. Never from your current location.
2. When the Moon appears to completely cover the Sun (an eclipse), the Moon must be at which phase?
 - A. Full
 - B. New
 - C. First quarter
 - D. Last quarter
 - E. At no particular phase
3. Imagine that you are building a scale model of the Earth and the Moon. You are going to use a 12-inch basketball to represent the Earth and a 3-inch tennis ball to represent the Moon. To maintain the proper distance scale, about how far from the surface of the basketball should the tennis ball be placed?
 - A. 4 inches (1/3 foot)
 - B. 6 inches (1/2 foot)
 - C. 36 inches (3 feet)
 - D. 30 feet
 - E. 300 feet
4. You have two balls of equal size and smoothness, and you can ignore air resistance. One is heavy, the other much lighter. You hold one in each hand at the same height above the ground. You release them at the same time. What will happen?
 - A. The heavier one will hit the ground first.
 - B. They will hit the ground at the same time.
 - C. The lighter one will hit the ground first.
5. How does the speed of radio waves compare to the speed of visible light?
 - A. Radio waves are much slower.
 - B. They both travel at the same speed.
 - C. Radio waves are much faster.
6. Astronauts inside the International Space Station (ISS) float around as it orbits the Earth because
 - A. there is no gravity in space.
 - B. they are falling in the same way as the ISS.
 - C. they are above the Earth's atmosphere.
 - D. there is less gravity inside the ISS.
 - E. more than one of the above.

PLEASE TURN THE PAGE

7. Imagine that the Earth's orbit were changed to be a perfect circle about the Sun so that the distance to the Sun never changed. How would this affect the seasons?
- We would no longer experience a difference between the seasons.
 - We would still experience seasons, but the difference would be much LESS noticeable.
 - We would still experience seasons, but the difference would be much MORE noticeable.
 - We would continue to experience seasons in the same way we do now.
8. Where does the Sun's energy come from?
- The combining of light elements into heavier elements
 - The breaking apart of heavy elements into lighter ones
 - The glow from molten rocks
 - Heat left over from the Big Bang
9. On about September 22, the Sun sets directly to the west as shown on the diagram below. Where would the Sun appear to set two weeks later?
- Farther south
 - In the same place
 - Farther north



10. If you could see stars during the day, this is what the sky would look like at noon on a given day. The Sun is near the stars of the constellation Gemini. Near which constellation would you expect the Sun to be located at sunset?
- Leo
 - Cancer
 - Gemini
 - Taurus
 - Pisces

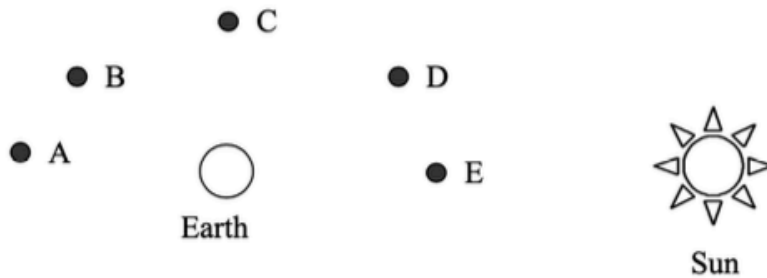


PLEASE TURN THE PAGE

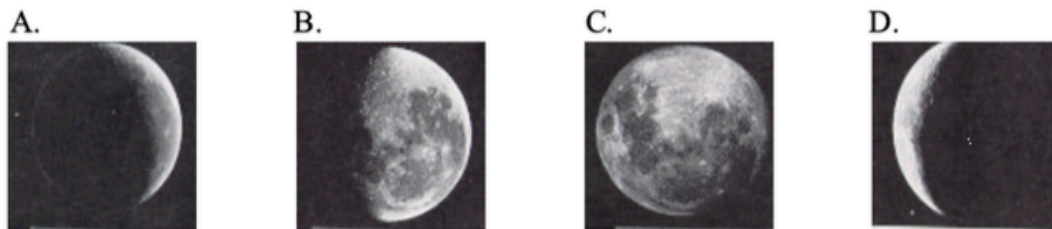
11. Compared to the distance to the Moon, how far away is the International Space Station from Earth?
A. Very close to the Earth
B. About half way to the Moon
C. Very close to the Moon
D. About twice as far as the Moon
12. As viewed from our location, the stars of the Big Dipper can be connected with imaginary lines to form the shape of a pot with a curved handle. To where would you have to travel to first observe a considerable change in the shape formed by these stars?
A. Across the country D. Moon
B. A distant star E. Pluto
C. Europe
13. Which of the following lists is correctly arranged in order of closest-to-most-distant from the Earth?
A. Stars, Moon, Sun, Pluto D. Moon, Sun, stars, Pluto
B. Sun, Moon, Pluto, stars E. Moon, Pluto, Sun, stars
C. Moon, Sun, Pluto, stars
14. Which of the following would make you weigh half as much as you do right now?
A. Take away half of the Earth's atmosphere.
B. Double the distance between the Sun and the Earth.
C. Make the Earth spin half as fast.
D. Take away half of the Earth's mass.
E. More than one of the above
15. A person is reading a newspaper while standing 5 feet away from a table that has on it an unshaded 100-watt light bulb. Imagine that the table were moved to a distance of 10 feet. How many light bulbs in total would have to be placed on the table to light up the newspaper to the same amount of brightness as before?
A. One bulb. D. Four bulbs.
B. Two bulbs. E. More than four bulbs.
C. Three bulbs.
16. According to modern ideas and observations, what can be said about the location of the center of the Universe?
A. The Earth is at the center.
B. The Sun is at the center.
C. The Milky Way Galaxy is at the center.
D. An unknown, distant galaxy is at the center.
E. The Universe does not have a center.
17. The hottest stars are what color?
A. Blue C. Red E. Yellow
B. Orange D. White

PLEASE TURN THE PAGE

18. The diagram below shows the Earth and Sun as well as five different possible positions for the Moon. Which position of the Moon would cause it to appear like the picture at right when viewed from Earth?



19. You observe a full Moon rising in the east. How will it appear in six hours?



Images © UC Regents/Lick Observatory. Unauthorized use prohibited.

20. With your arm held straight, your thumb is just wide enough to cover up the Sun. If you were on Saturn, which is 10 times farther from the Sun than the Earth is, what object could you use to just cover up the Sun?
- A. Your wrist
 - B. Your thumb
 - C. A pencil
 - D. A strand of spaghetti
 - E. A hair
21. Global warming is thought to be caused by the
- A. destruction of the ozone layer.
 - B. trapping of heat by nitrogen.
 - C. addition of carbon dioxide.
22. In general, how confident are you that your answers to this survey are correct?
- A. Not at all confident (just guessing)
 - B. Not very confident
 - C. Not sure
 - D. Confident
 - E. Very confident

PLEASE TURN THE PAGE

23. What is your college major (or current area of interest if undecided)?
- A. Business
 - B. Education
 - C. Humanities, Social Sciences, or the Arts
 - D. Science, Engineering, or Architecture
 - E. Other
24. What was the last math class you completed prior to taking this course?
- A. Algebra
 - B. Statistics
 - C. Trigonometry
 - D. Pre-Calculus
 - E. Calculus
25. What is your age?
- A. 0-20 years old
 - B. 21-23 years old
 - C. 24-30 years old
 - D. 31 or older
 - E. Decline to answer
26. Which best describes your home community (where you attended high school)?
- A. Rural
 - B. Small town
 - C. Suburban
 - D. Urban
 - E. Not in the USA
27. What is your gender?
- A. Female
 - B. Male
 - C. Decline to answer
28. Which best describes your ethnic background?
- A. African-American
 - B. Asian-American
 - C. Native-American
 - D. Hispanic-American
 - E. None of the above (see question 29 below)
29. Which best describes your ethnic background?
- A. African (not American)
 - B. Asian (not American)
 - C. White, non-Hispanic
 - D. Multicultural
 - E. None of the above (see question 28 above)

PLEASE TURN THE PAGE

30. How good at math are you?
- A. Very poor
 - B. Poor
 - C. Average
 - D. Good
 - E. Very good
31. How good at science are you?
- A. Very poor
 - B. Poor
 - C. Average
 - D. Good
 - E. Very good
32. Which best describes the level of difficulty you expect/experienced from this course?
- A. Extremely difficult for me
 - B. Difficult for me
 - C. Unsure
 - D. Easy for me
 - E. Very easy for me
33. How many astronomy courses at the college level have you taken?
- A. I'm re-taking this course.
 - B. This is my first college-level astronomy course.
 - C. This is my second college-level astronomy course.
 - D. I've completed more than two other college-level astronomy courses.

END OF SURVEY